Chunking Strategies Evaluation Framework

Evidence-Based Guidance for Choosing a RAG Chunking Strategy — with Pricing & Benchmark Data

Prepared for: Team AI

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Success criterion: *"For this type of data and use case, which chunking strategy should we choose, and what evidence supports that decision?"*

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1. Problem Statement & Scope

Chunking directly affects RAG retrieval accuracy, context preservation, latency, and cost, but no single strategy is universally best. This framework compares six core strategies — Fixed-Size, Recursive, Semantic, Sliding Window, Structure-Based, and Hierarchical — plus three upgrade paths — Contextual Retrieval, Late Chunking, and Adaptive/Agentic chunking — scored on retrieval accuracy, context preservation, response quality, latency, token usage, and cost, to answer one question directly: which strategy fits which data type and use case, and what evidence and cost support that choice.

Deliverables: a working comparison of each strategy against a common evaluation lens, the metrics needed to judge them (retrieval accuracy, context preservation, response quality, latency, token usage), and a decision framework mapping data types and business problems to a recommended strategy with supporting evidence.

2. Executive Summary — Key Findings

Finding	Data Point	Source
Chunking strategy outweighs model choice	Up to a 9-point recall gap, best vs. worst method, same corpus/retriever	Weaviate, Sept 2025
Recursive chunking is the strongest cheap default	69% end-to-end accuracy, #1 of 7 strategies tested	Vecta/FloTorch (vendor), Feb 2026
Semantic chunking: high recall, weaker answers	91.9% retrieval recall vs. only 54% end-to-end accuracy (~43-token fragments)	Chroma Research / FloTorch, 2026
The universal overlap rule is unsupported	No measurable benefit; only added indexing cost, tested with SPLADE retrieval	Bennani & Moslonka, arXiv:2601.14123, Jan 2026
Contextual Retrieval cuts retrieval failures	Up to 67% fewer top-20 failures with BM25 + reranking (5.7% → 1.9%)	Anthropic, Sept 2024
Adaptive, metric-guided chunking beats one-size-fits-all	Answer correctness 72% (from 62–64%); +30% more questions answered correctly (65 vs. 49)	de Moura Júnior et al., arXiv:2603.25333, LREC 2026
Structure-aware chunking wins on rigid-structure domains	87% vs. 13% accuracy against a fixed-size baseline, clinical decision support	MDPI Bioengineering, Nov 2025

> *Recommendation: start with recursive chunking (~512 tokens, no overlap by default), measure on your own corpus, and graduate to advanced strategies only when your own metrics justify the added cost.*

3. Strategy Comparison — Accuracy, Speed & Cost Multiplier

Strategy	End-to-End Accuracy	Retrieval Recall	Indexing Speed	Cost vs. Fixed-Size
Fixed-size	58–67%	76%	Fastest (baseline)	1×

Strategy	End-to-End Accuracy	Retrieval Recall	Indexing Speed	Cost vs. Fixed-Size
Recursive	69% (best)	88%	Fastest	1×
Sentence-based	~65% (est.)	Matches semantic to ~5k tok	Fast	~1×
Sliding window	Variable (high redundancy)	High (redundant hits)	Fast–moderate	1.5–3× (storage)
Structure-based (page-level)	0.648 avg. accuracy, lowest variance	High on structured docs	Fast	0.5–1× (fewer, larger chunks)
Semantic	54%	91.9% (best)	~14× slower	Higher (per-sentence embed calls)
Hierarchical (parent-child)	High (qualitative)	High	Moderate	~2× (dual vector sets)
Contextual Retrieval	Up to 67% fewer failures	N/A (pre-embedding step)	Slow preprocessing	See §4.2 pricing
Late chunking	High on long docs	BEIR gains grow with length	Moderate	Embedding-only (no LLM)
Adaptive / agentic	72% (from 62–64%), highest ceiling	Task-dependent	Slowest	10–50×

Sources: NVIDIA, June 2025 (page-level 0.648 accuracy, five-dataset average); Vecta/FloTorch (vendor), Feb 2026; Chroma Research; Anthropic, Sept 2024; Chonkie benchmark (semantic ~14× slower); Vectara, NAACL 2025 (arXiv:2410.13070); de Moura Júnior et al., arXiv:2603.25333, LREC 2026. Fixed-size = 1× baseline for cost comparison; treat vendor-produced benchmarks as directional.

3.1 Spotlight: Adaptive Chunking's Intrinsic Metrics

A March 2026 paper (accepted at LREC 2026) reframes chunking as something to score before it ever reaches a retriever. Instead of picking one chunker for an entire corpus, the Adaptive Chunking framework scores each document on five intrinsic, document-based metrics and selects the best-fitting chunking method per document, with no change to the downstream model or prompts:

- References Completeness (RC) — whether a chunk keeps a reference (e.g., “as shown above”) together with what it refers to.
- Intrachunk Cohesion (ICC) — how semantically unified the content within a single chunk is.
- Document Contextual Coherence (DCC) — how well a chunk preserves its relationship to the surrounding document.
- Block Integrity (BI) — whether structural units (tables, list items, code blocks) are kept intact rather than split.
- Size Compliance (SC) — how closely chunk sizes track the target range without over- or under-shooting.

On a corpus spanning legal, technical, and social science documents, metric-guided selection raised answer correctness to 72% (from a 62–64% baseline across the fixed strategies tested) and increased the number of correctly answered questions by over 30% (65 vs. 49) — without changing the underlying model or prompts.

The trade-off is cost: this approach sits at the "slowest / 10–50×" end of the comparison table above, so it is best reserved for smaller, high-value, structurally diverse corpora rather than applied wholesale.

Source: de Moura Júnior, Lelong & Blangero, "Adaptive Chunking: Optimizing Chunking-Method Selection for RAG," arXiv:2603.25333, March 2026, accepted at LREC 2026 (Ekimetrics).

4. Pricing Data (Verified August 2026)

Chunking strategy choice directly drives embedding and LLM API spend. The tables below use current, publicly listed vendor rates as of late August 2026. Confirm against each provider's pricing page before budgeting, since rates change frequently — the Claude Sonnet 5 row below is a concrete example of a rate scheduled to change within days of this framework's publication.

4.1 Embedding Model Pricing (per 1M input tokens)

Model	Standard Price	Batch Price (50% off)	Dimensions	Notes
OpenAI text-embedding-3-small	$0.02	$0.01	1,536 (reducible)	Best price-to-performance; industry default
OpenAI text-embedding-3-large	$0.13	$0.065	3,072 (reducible)	6.5× the cost of 3-small; marginal quality gain for most tasks
OpenAI text-embedding-ada-002 (legacy)	$0.10	$0.05	1,536	Superseded by 3-small; no reason to use for new projects
Google text-embedding-005	~$0.00625	N/A	768	Cheapest hosted option, ~3× under OpenAI 3-small
Jina Embeddings v3	~$0.02	N/A	1,024 (Matryoshka)	Supports native Late Chunking; long-context (8K tok)
Voyage-3-large	~$0.18	N/A	1,024–2,048	Highest MTEB retrieval quality; ~9× OpenAI 3-small cost

Sources: OpenAI pricing documentation; vendor pricing pages for Google, Jina, and Voyage, cross-checked late August 2026. Embeddings are input-only (no output-token cost).

4.2 Claude Model Pricing (for Contextual Retrieval preprocessing)

Model	Input ($/1M tok)	Output ($/1M tok)	Batch (50% off)	Cached Input (90% off)
Claude Haiku 4.5	$1.00	$5.00	$0.50 / $2.50	$0.10
Claude Sonnet 5	$2.00 (introductory, through Aug 31, 2026)	$10.00 (introductory)	$1.00 / $5.00	$0.20
Claude Opus 4.8	$5.00	$25.00	$2.50 / $12.50	$0.50

> *Timing note: Claude Sonnet 5's $2 / $10 rate is introductory and reverts to a standard $3 / $15 per million tokens on September 1, 2026 — four days after this framework's publication date. Re-check Anthropic's pricing page before budgeting a production Contextual Retrieval run on Sonnet 5.*

Contextual Retrieval typically uses the cheapest capable model (Haiku-class) to generate a 50–100 token context summary per chunk. Anthropic's original September 2024 write-up (using Claude 3 Haiku) reported approximately $1.02 per million document tokens preprocessed with prompt caching — re-verify against current per-token rates above before budgeting a production run, since both list prices and the specific model used have since changed.

Source: Anthropic API pricing, cross-checked against third-party rate trackers, late August 2026.

4.3 Vector Database Storage Cost at 10M Vectors (1,536-dim)

Database	Pricing Model	Approx. Monthly Cost	Notes
Pinecone Serverless	Storage + Read/Write Units	~$70–99/mo	Storage ~$0.30–0.33/GB; reads $8.25–16/M RU; writes $2–4/M WU; $50/mo plan minimum
Qdrant Cloud	Per-node (RAM/vCPU)	~$65/mo	Cheapest managed dedicated option at this scale; self-hosted is free (infra cost only)
Weaviate Cloud	Activity Units + storage tier	~$135/mo	No quantization; Binary Quantization can cut cost ~5×
pgvector (self-hosted on RDS)	Postgres instance cost	~$45–180/mo	Effectively free beyond the Postgres instance you already run

Sources: independent vector-database pricing trackers, cross-checked June–August 2026. Costs scale with vector count, dimensionality, and query volume — treat as directional, not a quote; confirm against Pinecone, Qdrant, and Weaviate's own pricing pages before budgeting.

5. Overlap and Chunk Size Guidance

Retrieval Method	Use Overlap?	Recommended Amount
Dense-only (vector similarity)	Yes	50–100 tokens (10–20% of chunk size)
Sparse-only (BM25 / SPLADE)	No	0 — no measurable benefit found, per arXiv:2601.14123 (Jan 2026)
Hybrid (dense + sparse)	Yes	~50 tokens
With reranker	Yes	~50 tokens

Query Type	Optimal Chunk Size	Recommended Strategy
Factoid / lookup	64–256 tokens	Sentence or sentence-window
Analytical / reasoning	512–1,024 tokens	Recursive or hierarchical
Summarization	1,024+ tokens	Hierarchical (small-to-big)
Code / technical reference	Function / block level	AST-aware code splitter
Legal / contract	Clause-aligned	Adaptive / boundary-aware
Long narrative	512–1,024 tokens	Late chunking

Context cliff: the same January 2026 preprint found response quality degrading beyond roughly 2,500 tokens per chunk, and found sentence-based chunking matched semantic chunking up to ~5,000 tokens at a fraction of the cost — directional findings from a single preprint, not settled constants, but a useful ceiling and a caution against reaching for semantic chunking by default.

6. Evaluation Criteria

Dimension	Metrics	Target / Threshold
Retrieval quality	Recall@k, Precision@k, MRR, nDCG	Beat the fixed-size baseline by a measured margin before adopting
End-to-end quality	RAGAS context precision/recall, faithfulness, answer relevancy	LLM-as-judge score improves without a >2× cost increase
Speed & cost	Indexing time, cost multiplier, query latency (p50/p95)	Stay within the agreed cost-per-1K-queries budget
Context cliff avoidance	Average and max chunk token count	Keep chunks well below ~2,500 tokens
Domain fit	Accuracy on a domain-specific held-out set	Structure-aware wins required for legal, clinical, and technical domains
Maintenance overhead	Re-indexing frequency, prompt/threshold drift	Document any recurring tuning cost

7. Decision Framework

1. **Baseline —** recursive chunking, ~512 tokens, no overlap by default.
2. **Match size to query type —** see the query-type table in §5.
3. **Measure —** recall@k, precision@k, and answer quality separately on a held-out set.
4. **Respect structure —** if documents have headings, code, or pages, switch to structure-based chunking — usually a low-risk win.
5. **Diagnose —** answers spanning chunks → increase size/overlap first; thin context → hierarchical; broken cross-references → contextual retrieval or late chunking.

6. **Earn the complexity —** adopt semantic, hierarchical, contextual, late, or adaptive chunking only if your own metrics justify the added cost (see §4 pricing).
7. **Document —** metrics, configuration, and cost for reproducibility.

7.1 Decision Matrix by Use Case

Document Type	Default Strategy	Upgrade If Needed	Est. Cost Driver
General articles / reports	Recursive, 512 tok	Semantic (proven recall gap)	Embedding tokens only
FAQs / short support docs	No chunking / sentence-level	Header-aware for large KBs	Minimal
Markdown / HTML / code	Header- or AST-aware + recursive	Hierarchical for deep nesting	~1× embedding, +2× vectors if hierarchical
PDFs / financial reports	Page-level / layout-aware	Structure-based for dense tables	Parsing compute (OCR/layout)
Legal contracts / policies	Hierarchical (parent-child)	Contextual retrieval for cross-refs	+LLM cost (Haiku-class, §4.2)
Transcripts / logs	Sliding window / sentence-based	Semantic, 200–400 tok floor	Higher vector count (redundancy)
Long cross-referential manuals	Late chunking	+ Hierarchical parent-child retrieval	Long-context embedding model
High-value, unstructured, low-volume	Recursive baseline first	Adaptive / LLM-based (agentic)	10–50× (LLM calls per doc)

8. Practical Recommendation

Recursive chunking at ~512 tokens remains the safest, cheapest, and most consistently accurate starting point across every benchmark reviewed here — at $0.02/M tokens (OpenAI 3-small, batch: $0.01/M) it also carries the lowest embedding cost of any strategy in this framework. Advanced strategies should be adopted only when Team AI's own evaluation shows a measurable accuracy gain that outweighs the added latency and the incremental LLM/vector-DB cost quantified in §4.

This avoids two traps at once: the “overlap myth” of copying a 10–20% overlap default that current evidence does not support for sparse retrieval, and the habit of chasing retrieval-only metrics (like semantic chunking's 91.9% recall) that don't reliably translate into better end-to-end answers or justify their price. Where documents have genuine structure — legal clauses, technical manuals, clinical protocols — the evidence is equally clear in the other direction: structure-aware and adaptive methods produce the largest, most defensible gains in this review (87% vs. 13% in clinical decision support; 72% vs. 62–64% with metric-guided adaptive chunking), and are worth the added cost precisely because the fixed-size baseline is unusually weak on that kind of document.

9. References

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